

Environment Setup Guide

for
DSRC V2X Product

Revision: 1.0

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1. Objective

The purpose of this document is to give a stepwise illustration about how to setup a development environment for Unex DSRC V2X products, including both OBU and RSU, and both IEEE 1609.x and ETSI TC-ITS protocol stack

A building image created by the development environment can be flash into the supported Unex V2X products.

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2. Reference

» Autotalks SDK User Manual / 4.11.1

» Autotalks SDK User Manual / 4.10.3

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3. Scope

Supported product:

» OBU-201U

» RSU-101U

» OBU-201E

» RSU-101E

Supported protocol stack:

» Unex WAVE-SDK-v1007 or above (apply Autotalk SDK-4.11.1)

» Unex EU-SDK-v1028 or above (apply Autotalk SDK-4.10.3)

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4. Environment Setup Step

This chapter illustrates the step-by-step procedure to setup the development environment, and the corresponding reference CLI commands under Linux environment. Please note that the reference CLI commands might be varied due to different environment

4.1. Setup a Linux Environment

There are two types of development host machine recommended:

» Ubuntu/Xubuntu Linux 12.04 LTS 32-bit OS

» Ubuntu/Xubuntu Linux 14.04 LTS 64-bit OS.

Please download the dedicated version from Ubuntu official website

You can setup a native or a virtual machine of Linux environment. No matter native or virtual, be sure that internet access is working.

For 64-bit OS, install additional package as following:

```
sudo dpkg --add-architecture i386
sudo apt-get install libc6:i386 lib32z1
```

4.2. Installing GCC ARM cross-compiler

The GCC AMR cross-compiler packed in the tar file "`gcc-arm-none-eabi-4_8-2014q3-atk-1.0.0-linux.tar.xz`" in the released SDK. Please download it to your home directory in the Linux environment. The following commands install the

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GCC ARM cross-compiler:

```
tar -xf gcc-arm-none-eabi-4_8-2014q3-atk-1.0.0-linux.tar.xz
sudo mkdir -p /tools/gcc/arm
sudo mv gcc-arm-none-eabi-4_8-2014q3-atk-1.0.0 /tools/gcc/arm
rm gcc-arm-none-eabi-4_8-2014q3-atk-1.0.0-linux.tar.xz
```

4.3. Installing Autotalk SDK

» For WAVE-SDK, apply Autotalk SDK-4.11.1 (`sdk-4.11.1-sc-i686-linux-gnu.tar.xz`).

» For EU-SDK, apply Autotalk SDK-4.10.3 (`sdk-4.10.3-sc-i686-linux-gnu.tar.xz`).

The tar file included in the Unex released SDK. Please download the tar file to your home directory in the Linux environment.

4.3.1. For WAVE-SDK

The following commands install Autotalk SDK-4.11.1

```
tar -xf sdk-4.11.1-sc-i686-linux-gnu.tar.xz
sudo mkdir -p /tools/autotalks
sudo mv sdk-4.11.1-sc-i686-linux-gnu /tools/autotalks
rm sdk-4.11.1-sc-i686-linux-gnu.tar.xz
```

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4.3.2. For EU-SDK

The following commands install Autotalk SDK-4.10.3

```
tar -xf sdk-4.10.3-sc-i686-linux-gnu.tar.xz
sudo mkdir -p /tools/autotalks
sudo mv sdk-4.10.3-sc-i686-linux-gnu /tools/autotalks
rm sdk-4.10.3-sc-i686-linux-gnu.tar.xz
```

4.4. Build Example in Unex SDK

Make a temporary directory under your home directory. And download the Unex SDK into the directory and try to build the example. Following commands make a "sdk" directory in home directory:

```
mkdir ~/sdk
```

4.4.1. For WAVE-SDK

Download the WAVE-SDK package file into "sdk" folder. For example, "`sdk-us-1007-uC.tar.gz`". The "`1007`" represent the version number. Please note that the package filename might be different for different release version. Try following commands to build the example:

```
cd ~/sdk/
tar -xf sdk-us-1007-uC.tar.gz
rm sdk-us-1007-uC.tar.gz
cd sdk-us-1007-uC/example/test/
make clean; make
```

After make done, check "`./arm/img/`" with following command. There should be an image file "`wave-sdk.img`"

```
ls ./arm/img/
```

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4.4.2. For EU-SDK

Download the EU-SDK package file into “sdk” folder. For example, “`sdk-eu-1028.zip`”. The “`1028`” represent the version number. Please note that the package filename might be different for different release version. Try following commands to build the example:

```
cd ~/sdk/
unzip sdk-eu-1028.zip
rm sdk-eu-1028.zip
cd sdk-eu-1028/sdk/
make clean; make
```

After make done, check “`./arm/img/`” with following command. There should be an image file “`wave-sdk.img`”

```
ls ./arm/img/
```

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5. How to Build DSRC Applications?

DSRC applications can be built by modifying craton_user_init() function in main.c file, located under the sdk folder.

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